香港中文大學（深圳）
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PHY1001 Mechanics

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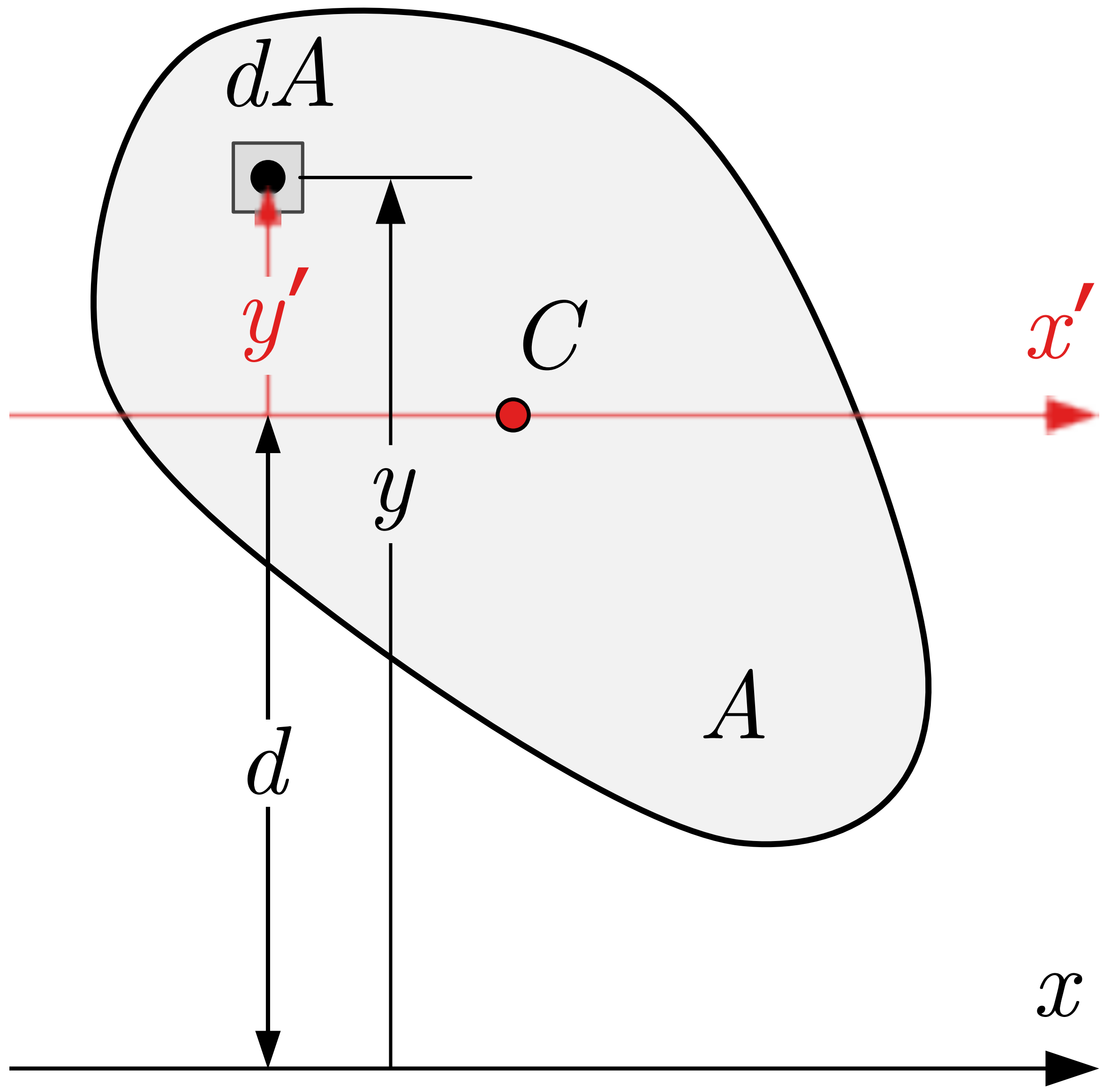
Parallel-Axis Theorem

$I = \int r^2 \, dm$

- A rigid body can have many moments of inertia depending on the axis.
- Parallel-axis theorem: $I_p = I_{cm} + Md^2$, where
  - I_p is the moment of inertia through axis p
  - I_{cm} is the moment of inertia through an axis that is through the center of mass and parallel to axis p
  - M is the total mass
  - d is the distance between the two axis

Proof of Parallel-Axis Theorem



axis$_c$ // axs P

$$\vec{r}_{cm} = \frac{\int \vec{r}\, dm}{M}$$

$$M \cdot \vec{r}_{cm} = \int y' dA = 0$$

$$I_p = \int r^2 dm$$

$$= \int y^2 dm$$

$$= \int (y' + d)^2 dA \cdot \rho$$

$$= \rho \int y'^2 dm + \rho \int d^2 dA + 2d\rho \int y' \cdot dA$$

$$= I_{cm} + md^2 + 0$$

$$\int y' dA = 0$$

$$\boxed{I_p = I_{cm} + md^2}$$

Exercise: A Thin Rod

A thin rod of mass M and length L rotates around an axis through its center and perpendicular to the rod. Find the moment of inertia I. Suppose the density is uniform.

(Hint: We know $I = \frac{1}{3}ML^2$ when the axis is through the end and perpendicular to the rod, the last exercise from the last lecture)

M1:

$$I = \int r^2 \, dm$$

$$= \int_{-\frac{L}{2}}^{\frac{L}{2}} l^2 \cdot \rho \, dl$$

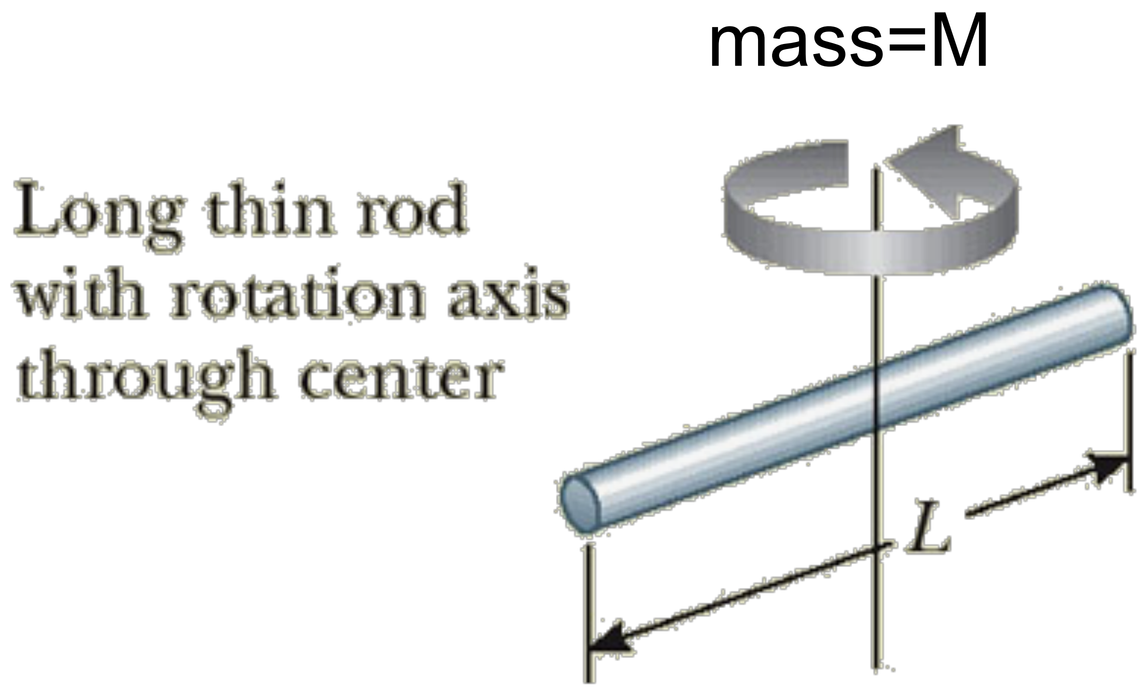
$$= \rho \frac{l^3}{3} \Big|_{-\frac{L}{2}}^{\frac{L}{2}}$$

$$= \rho \cdot \frac{1}{3} \cdot \frac{L^3}{8} \cdot 2 = \frac{\rho L^3}{12}$$

$m = \rho L$

$$\frac{I}{m} = \frac{L^3}{12}$$

$$\boxed{I = \frac{m}{12} L^2}$$



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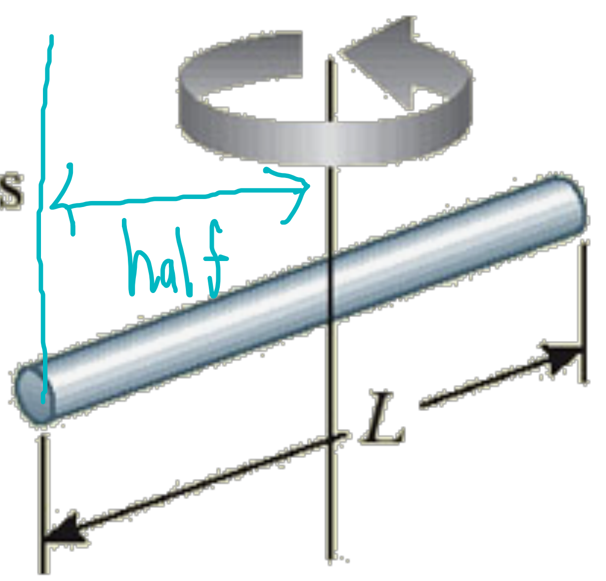
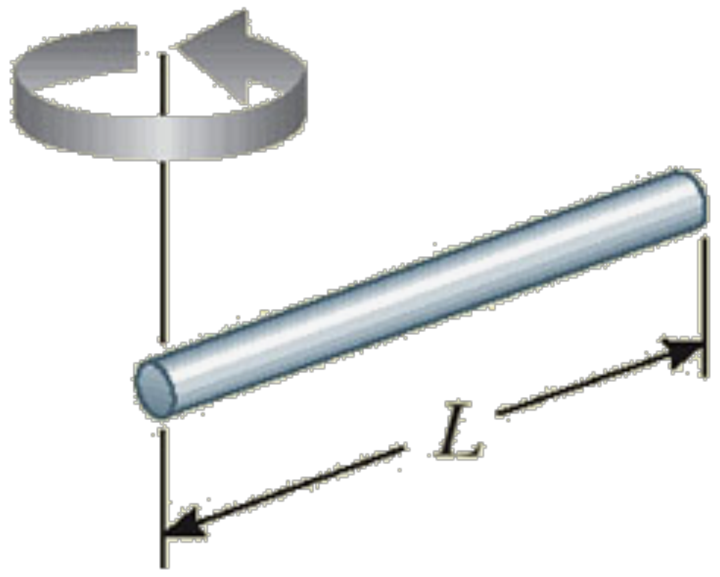
M2: $I_p = \frac{M}{3} L^2 = I_{cm} + M\left(\frac{L}{2}\right)^2$

$= I_{cm} + \frac{ML^2}{4}$

$I_{cm} = ML^2\left(\frac{1}{3} - \frac{1}{4}\right)$

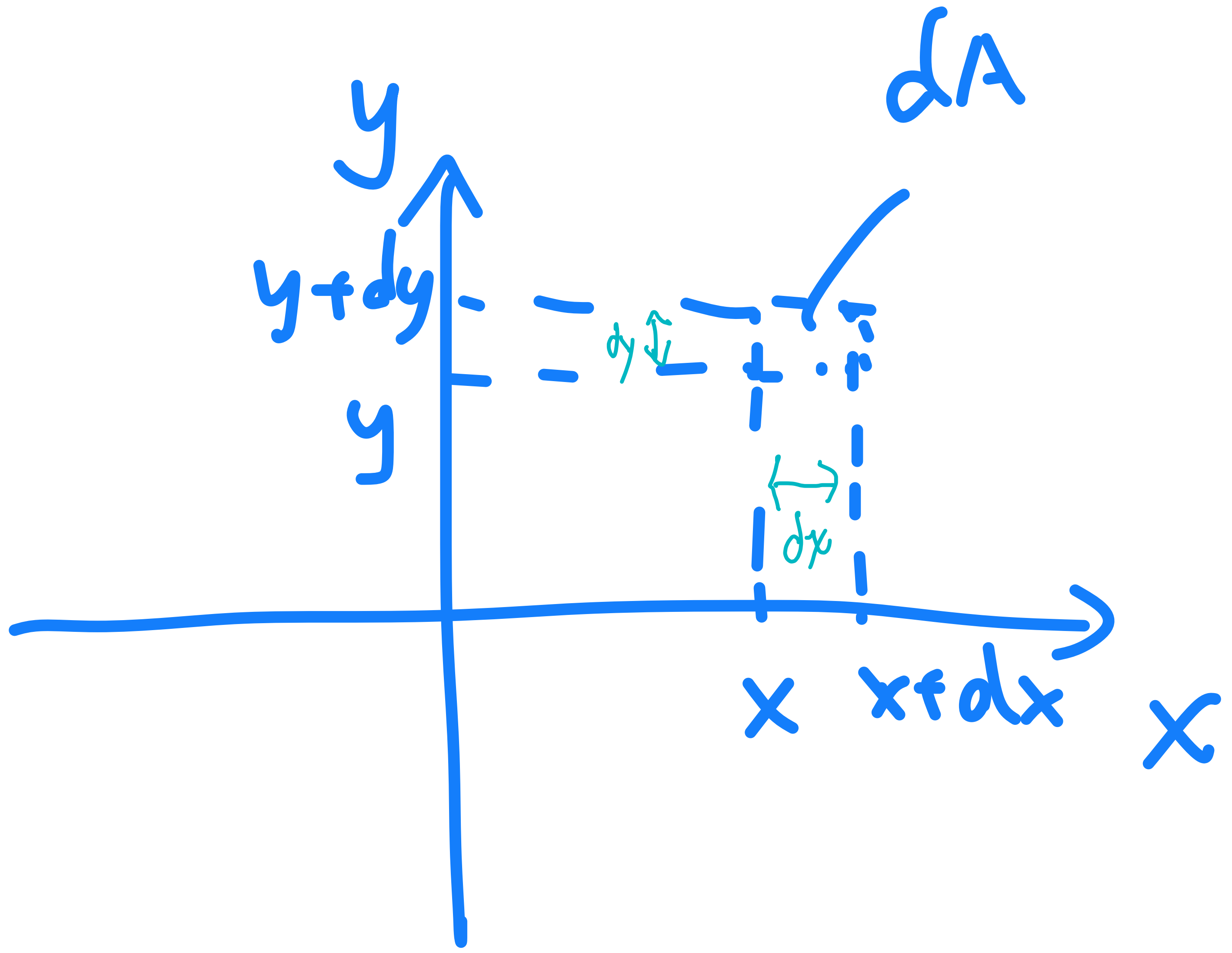
$= \frac{ML^2}{12}$





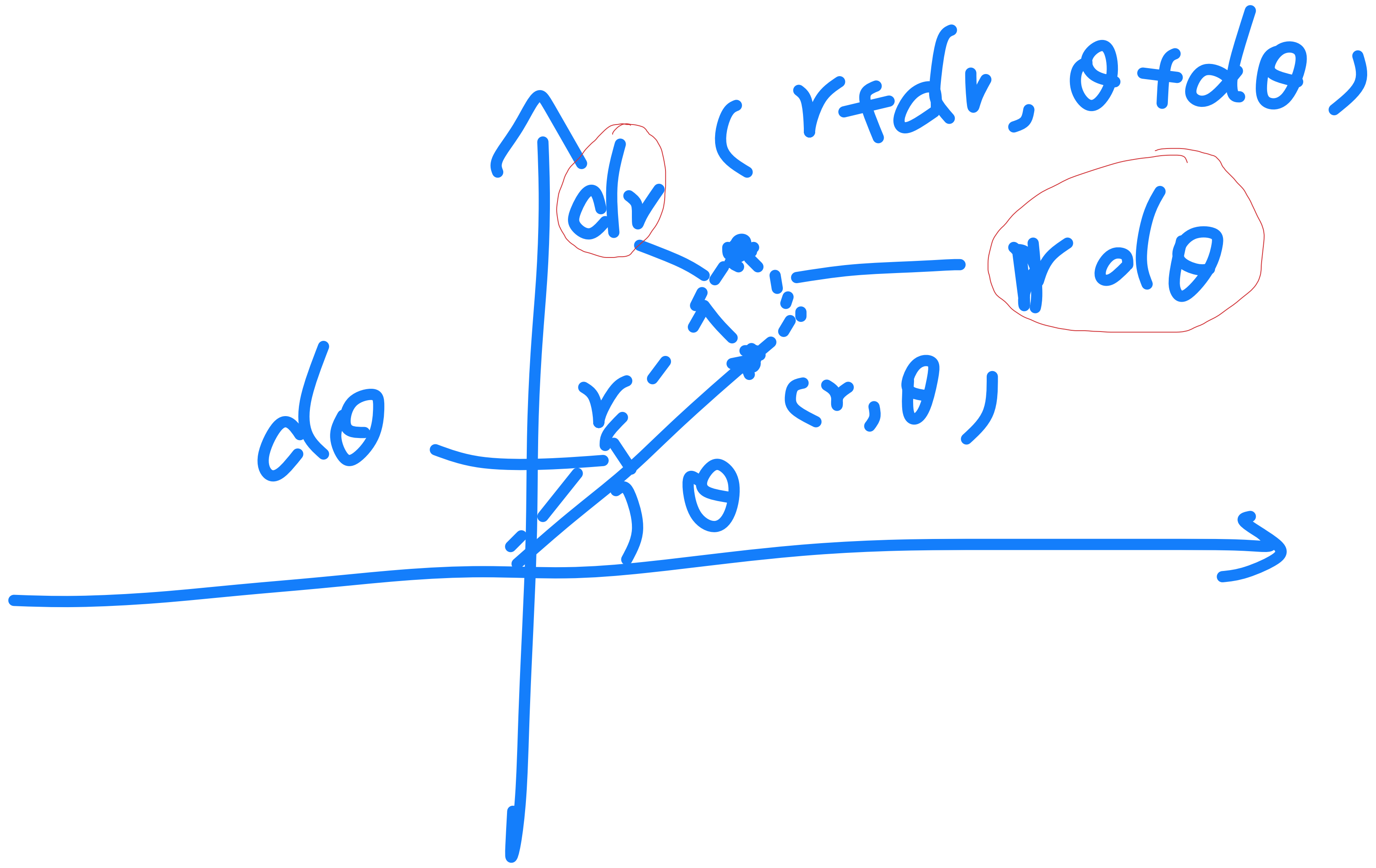
Polar Coordinates

Cartesian coordinate



$$x = r\cos\theta$$
$$y = r\sin\theta$$

Polar coordinate



$x : (-\infty, +\infty)$

$y : (-\infty, +\infty)$

$dA = dxdy$

Double integral

$$\int f(x, y)dA = \int f(x, y)\,dx\,dy = \int_{x_1}^{x_2}\int_{y_1}^{y_2} f(x, y)\,dx\,dy$$

$r : (0, +\infty)$

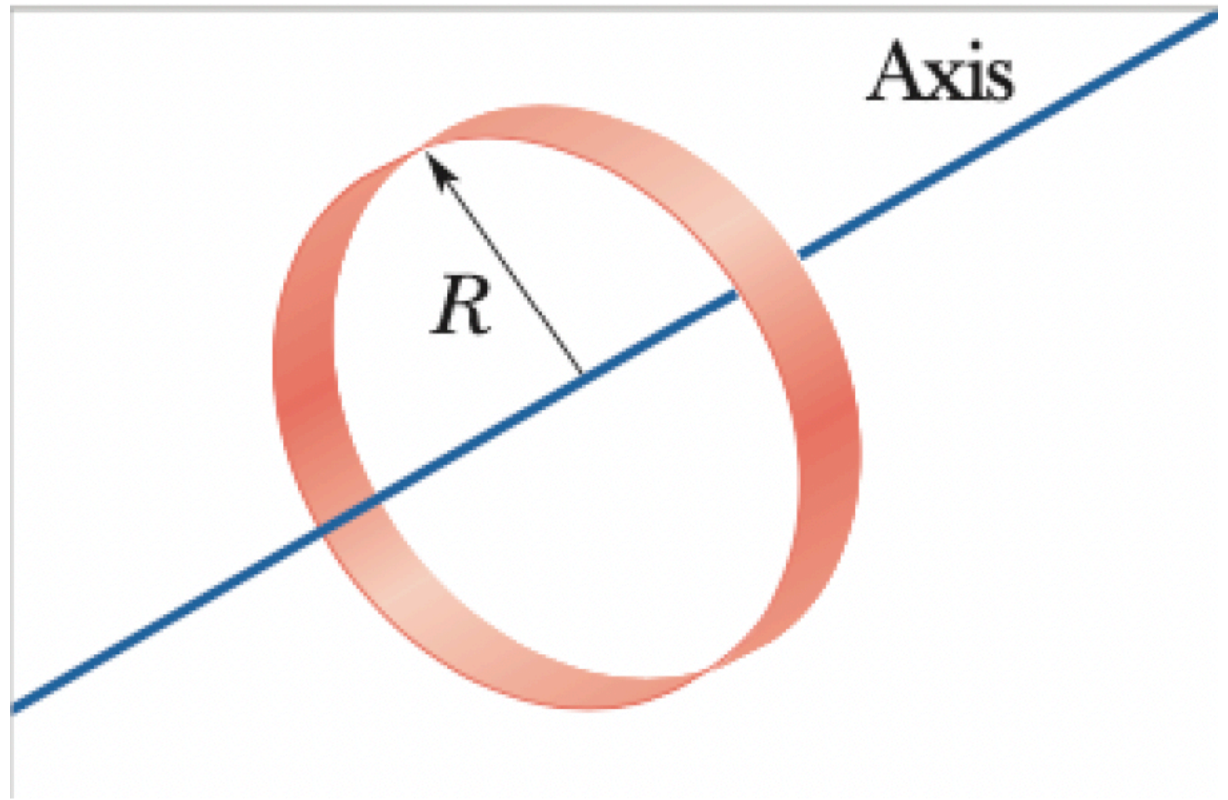
$\theta : (0, 2\pi)$

$dA = rdrd\theta$

$$dA = dr \cdot rd\theta = rdr\,d\theta$$

$$\int f(r, \theta)dA = \int f(r, \theta)\,r\,dr\,d\theta = \int_{\theta_1}^{\theta_2}\int_{r_1}^{r_2} f(r, \theta)\,r\,dr\,d\theta$$

Hollow Hoop



A hollow hoop of mass m and radius R rotates around an axis through its center and perpendicular to the hoop, as shown in the figure. Find the moment of inertia. Suppose the density is uniform.

$$\boxed{I = mR^2}$$

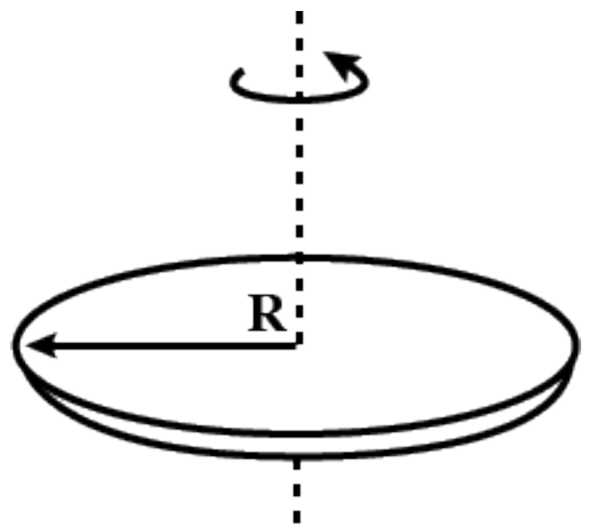
$$I = \int r^2 \, dm$$

$$= \int R^2 \, dm$$

$$= R^2 \int dm = R^2 m$$

→ disk = many layers of hollow hoops

Mass=m



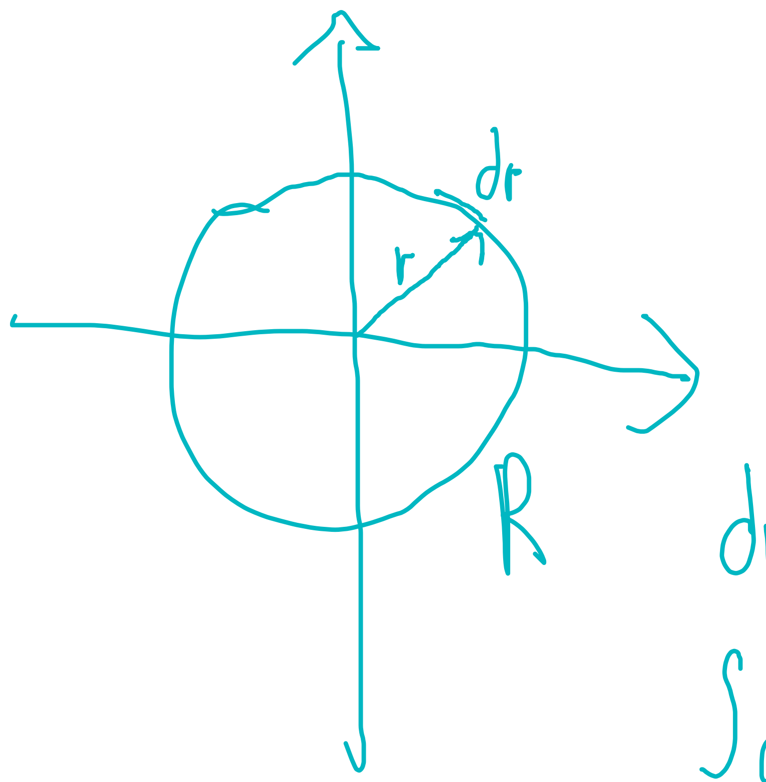
A disk of mass m and radius R rotates around an axis through its center and perpendicular to the disk, as shown in the figure. Find the moment of inertia. Suppose the density is uniform.

$$I = \int r^2 \, dm$$

$$m = \rho \pi R^2$$

$$\frac{I}{m} = \frac{R^2}{2}$$

$$I = \frac{mR^2}{2}$$



$f(r,\theta)$
$= g(r)$

$$= \int r^2 \cdot \rho \, dA$$

$$= \int r^2 \rho \cdot r \, dr \, d\theta$$

no θ term??

$$= \int_0^{2\pi} d\theta \int_0^R r^3 \rho \, r \, dr$$

$$= 2\pi \cdot \frac{\rho r^4}{4}\Big|_0^R = \frac{\pi \rho R^4}{2}$$

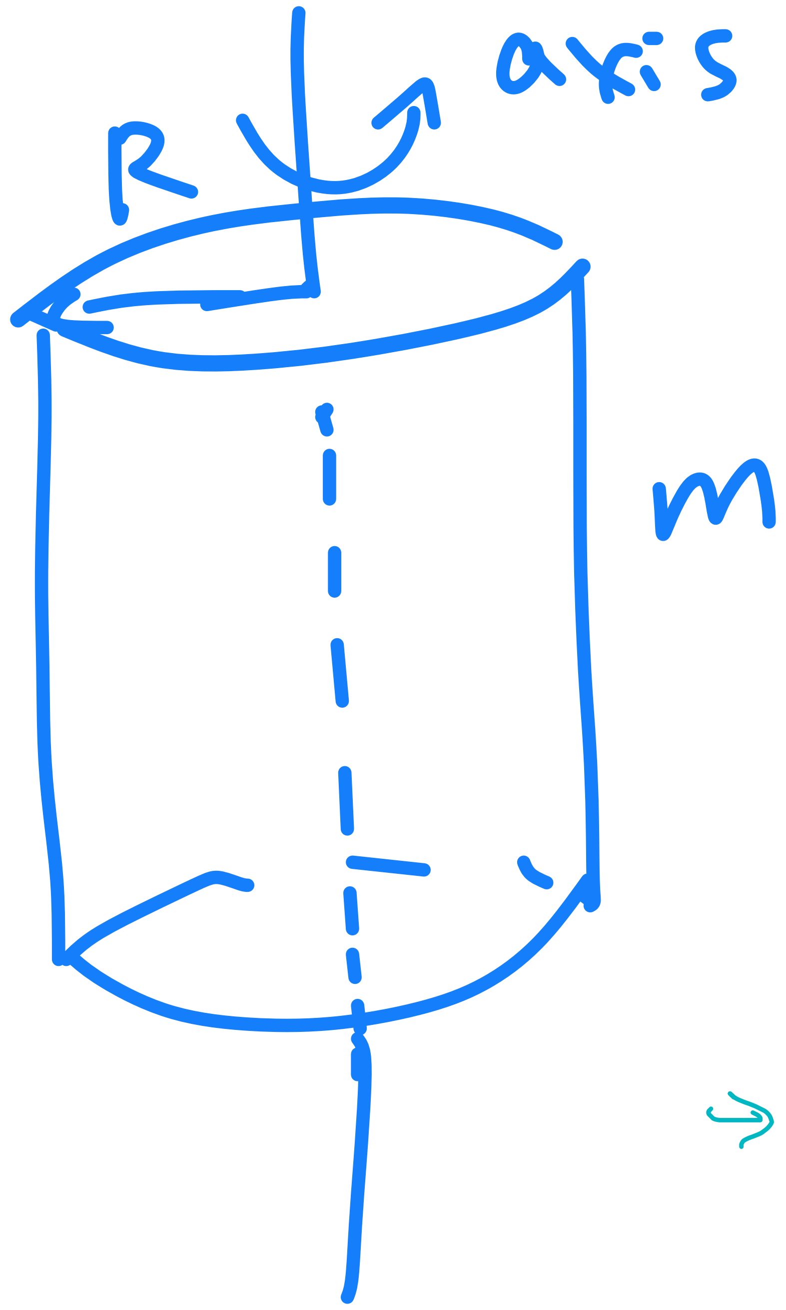
$dm = 2\pi r \cdot dr \cdot \rho$

$\int dI = \int dm \cdot r^2 = \int_0^R 2\pi r^3 dr \cdot \rho$

$= 2\pi \cdot \rho \cdot \frac{R^4}{4} = \frac{\rho \pi R^4}{2}$

$I = \frac{mR^2}{2}$

Cylinder



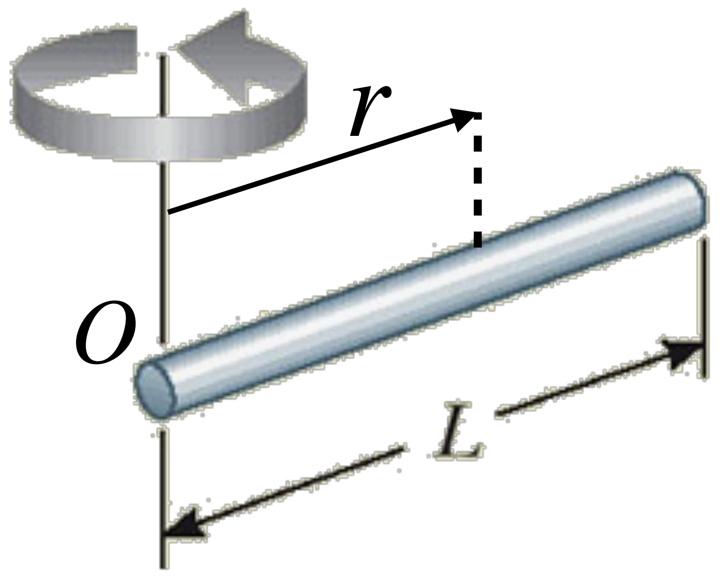
A cylinder of mass m and radius R rotates around an axis through its center and perpendicular to the cylinder, as shown in the figure. Find the moment of inertia. Suppose the density is uniform.

$$I = \frac{1}{2} m R^3$$

→ same as disk

Non-Uniform Rod

$\rho = \alpha r$



A rod of mass m and length L rotates around an axis through its end and perpendicular to the rod, as shown in the figure. The density $\rho = \alpha r$, where α is a positive constant, and r is the distance to the end. Find the moment of inertia in terms of m and L.

$$I = \int r^2 \, dm$$

$$= \int_0^L r^2 \cdot \alpha r \cdot dr$$

$$= \int_0^L \alpha r^3 \, dr$$

$$= \frac{\alpha r^4}{4} \Big|_0^L = \frac{\alpha L^4}{4}$$

$$m = \int_0^L \rho \, dl$$

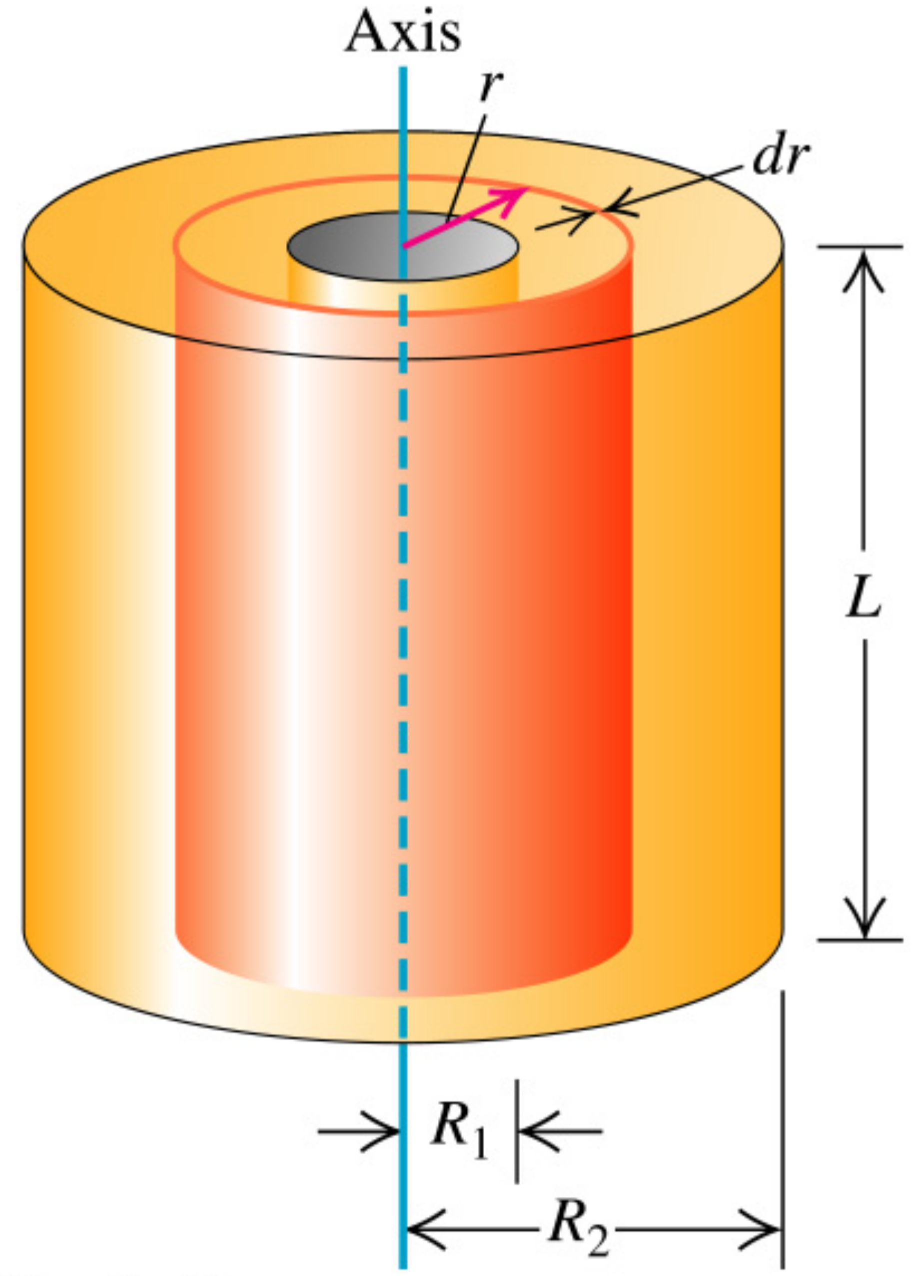
$$= \int_0^L \alpha r \, dr$$

$$= \frac{\alpha L^2}{2}$$

$$\frac{I}{m} = \frac{L^2}{2}$$

$$\boxed{I = \frac{mL^2}{2}}$$

Hollow Cylinder



A hollow of mass m rotates around an axis through its center and perpendicular to its horizontal surface, as shown in the figure. The inner and outer radius is R_1 and R_2, respectively. Find the moment of inertia. Suppose the density is uniform.

$$I = \int r^2 \, dm$$

$$= \int_0^{2\pi} \int_{R_1}^{R_2} r^2 \, r \, dr \, d\theta \, \rho$$

$$= 2\pi \cdot \int_{R_1}^{R_2} r^3 \rho \, dr$$

$$= \frac{\rho \pi}{2} (R_2^4 - R_1^4)$$

$\hookrightarrow (R_2^2 + R_1^2)(R_2^2 - R_1^2)$

$$m = \pi (R_2^2 - R_1^2) \cdot \rho$$

$$\frac{I}{m} = \frac{1}{2} (R_1^2 + R_2^2)$$

$$\boxed{I = \frac{m}{2} (R_1^2 + R_2^2)}$$

1. $R_1 \to 0$

$I = \frac{m}{2} R_2^2 \to$ solid cylinder

2. $R_1 = R_2$

$I = m R_2^2 \to$ hollow hoop